LFM3-1.7B: tri-hybrid layer stack and decode-state scaling

24 layers (config.json layer_types, cross-checked vs Liquid AI blog, 2026-10-27)

Layer sequence (bottom of stack first): 13 conv + 5 SWA + 6 global attn

C: short conv, k=7 (13) S: SWA, window 256 (5) G: global attn (6)

Every block: sandwich norm (in + out); attn layers add head-wise sigmoid gate

G: half RoPE, theta 1e6; S: full RoPE, theta 1e4; 16 Q / 4 KV heads

Decode state per layer type vs context length T

C: conv (13 layers)
fixed kernel-state buffer
(kernel size 7)
O(1): no KV cache
conv_L_cache = 7

S: SWA (5 layers)
Retain 256 tokens
if runtime crops cache
O(1): 2.5 MiB total
5 x 2 x 4 x 128 x 256 x 2B

G: global attn (6 layers)
full-context KV
(GQA: 4 KV heads x 128)
O(T): 12 KiB / token
6 x 2 x 4 x 128 x 2B

6 global layers: 12,288 B/token x 131,072 tokens = 1.611 GB (BF16)
Vendor reports 1.6 GB at 128K; actual runtime allocation not tested here.

Sources: Liquid AI blog (2026-10-27); HF config (retrieved 2026-10-28).
Independent explanatory diagram, not vendor artwork. Post-2026-08-30 supplemental material.